

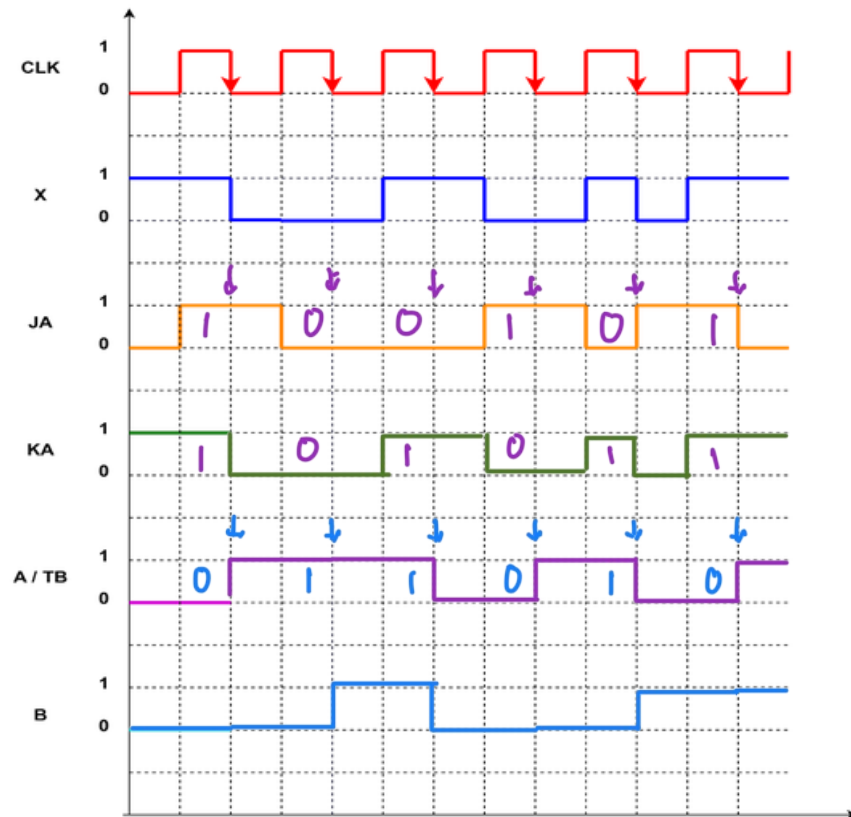
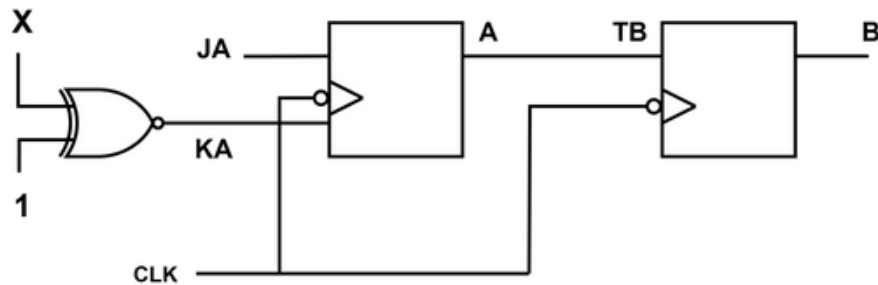
CSE260 Summer 2025

Assignment 4

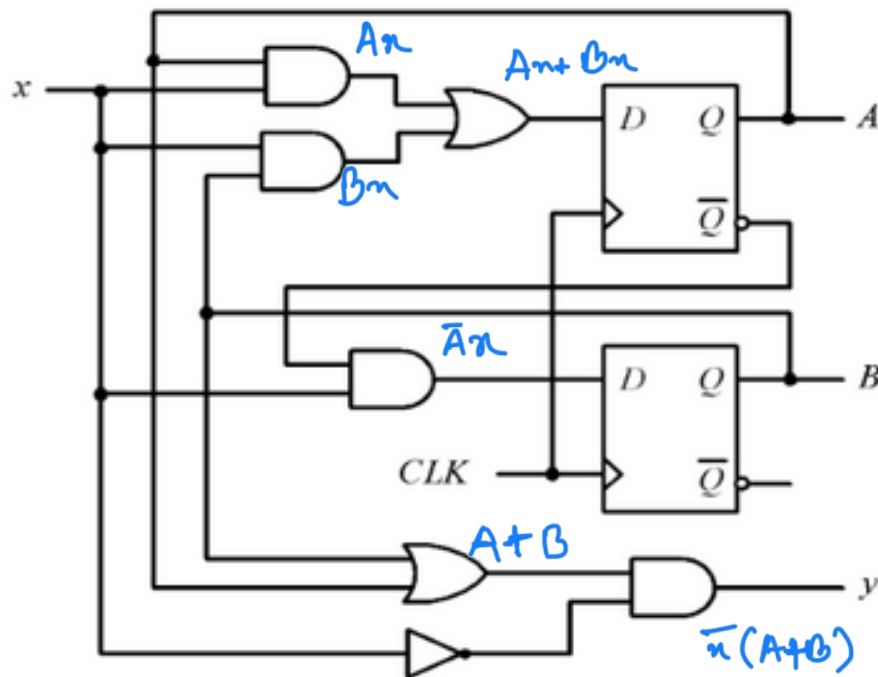
Total: 20

This assignment must be handwritten. Show ALL steps in ALL questions.

- [5 Marks] Draw and complete the **timing diagram** for the following circuit, where all the flip-flops are **negative-edge-triggered**. Clock Pulse(CLK), JA, and X waveforms are given.



2. [7 Marks] Draw the state diagram for the given circuit. Show all intermediate steps.



3. [8 Marks] Implement the following counter using **JK FF**. Show all steps.

Green->Orange->Yellow->Red->Yellow->Orange->Yellow->Green *Same as Quiz.*

2. $PS \rightarrow A, B; NS \rightarrow A^+, B^+$
Internal input = D_A, D_B
External input = n
Output = j

Exs

$D_A = A_n + B_n$ $y = \bar{a}(A+B)$

$D_B = \bar{A}n$

A	B	x	D _A	D _B	y	A ⁺	B ⁺
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	1	0	0	1	0

